

Henrique Baptista

AI Engineer | Agents · LLM Evaluation · Voice AI

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PROFESSIONAL SUMMARY

AI Engineer who builds production agent, LLM, and Voice AI systems, plus the evaluation layer that proves they work. Three years owning AI problems end to end across healthcare and SaaS: LLM-as-Judge evaluators running in production, multi-agent systems on LangGraph, ASR benchmarking with domain-specific metrics, and live products I design, ship, and operate myself.

PROFESSIONAL EXPERIENCE

ML / AI Engineer - VOA Health

Feb 2025 – Jul 2026

São Paulo, Brazil · HealthTech startup building AI for clinical documentation

The product generated clinical documents at scale, but quality was invisible: failures surfaced only when clinicians noticed them. My role was to make quality measurable and turn it into engineering decisions.

LLM Quality & Evaluation

- Shipped **LLM-as-Judge evaluators to production** (LangSmith online evaluators), scoring AI-generated clinical documents on faithfulness, correctness, and format adherence, so quality regressions were caught before clinicians ever saw them
- Calibrated those judges against human annotation using **Cohen's κ agreement gates** before rollout, and replaced ad-hoc review with a QA gate combining deterministic detectors with a root-cause bug catalog

Voice AI & ASR

- Benchmarked **9 commercial ASR providers on 10k+ clinical audio samples** (WER/CER, latency, cost) and created a medication-error metric for the domain, since a generic WER score never reveals whether a provider transcribes a drug name correctly
- Built a taxonomy of real medication transcription errors and a **lexical-phonetic post-processor** that corrects misrecognized drug names before they reach the note; the taxonomy drove provider selection, Portuguese normalization rules, and fine-tuning priorities. Tuned VAD to cut false segment breaks in noisy consultation audio

Data Curation & Observability

- Fine-tuning Whisper for clinical audio needed thousands of validated audio-text pairs, and manual exports could never scale. Built the full curation pipeline: audio chunking, Hugging Face Hub datasets, a custom **Argilla schema with dual-annotator consensus** and adjudication, a progress dashboard, and a 24/7 webhook service syncing approved annotations straight to the Hub. Instrumented the realtime pipeline end to end with Prometheus and Grafana, turning provider and endpoint selection into data-backed decisions

PROJECTS

WordinAI (solo-built, live at [wordinai.com](#))

LangGraph · deepagents · Next.js · Supabase

- SaaS that turns long-form audio into structured documents, built on a **multi-agent LangGraph system**: a context analyst, a source verifier that requires a reference for every claim, and an action strategist, working over hybrid retrieval (vector plus full-text) of the user's own material
- Operated end to end: multi-provider STT routing with retries and circuit breakers, in-browser Whisper WASM as a zero-cost privacy mode, subscription billing, multi-tenant Supabase RLS, test suites, and Sentry plus PostHog observability

Wilmec (shop-management system for an auto mechanic shop, live in production) OpenAI Agents SDK · Next.js · Supabase · Whisper

- Sole engineer** of the system a real repair shop runs on every day: work orders, scheduling, vehicle inspection, inventory, financials, and invoicing across **34 Postgres tables with 103 row-level security policies** and role-based access per job. Offline-first (IndexedDB plus a sync queue with cascading ID remapping) because the shop floor has no signal; migrated **2,078 clients and 5,527 historical work orders** off a 2018 Firebird database with an idempotent, record-validated pipeline
- Built the AI layer on the **OpenAI Agents SDK**: a conversational inventory agent with 6 Zod-typed tools serving **Telegram, WhatsApp, and web through one engine**, with tool-level guardrails (confirmation gate on anomalous stock movements, duplicate detection on creation), sender allowlist, rate limiting, and an audit trail from every stock movement back to the message that caused it
- Shipped **voice to structured data** (Groq Whisper → strict-JSON LLM → validated server action) so mechanics dictate work-order line items from the floor, plus predictive diagnostics assembled server-side from each vehicle's last 30 work orders and inspections, with explicit anti-hallucination rules and every AI interaction logged for later evaluation

Earlier ML & Computer Vision

XGBoost · FastAPI · Detectron2 · AWS

- Education Risk Prediction** (Datathon Passos Mágicos): early-warning system for student risk at an NGO, XGBoost classifier (**92.59% accuracy, 0.97 ROC-AUC**) behind a FastAPI service with a decision dashboard and drift monitoring, deployed on Hugging Face Spaces
- Insurance Claim Computer Vision** (Porto Seguro Challenge, **Top-10 Finalist**): Detectron2 segmentation pipeline with a COCO dataset built from scratch and transfer learning on AWS, cutting manual claim-review workload by roughly 80%

EDUCATION

Postgraduate Program, Machine Learning Engineering - FIAP

Completed Mar 2026

Bachelor, Systems Analysis & Development - FIAP

Completed Dec 2024

TECHNICAL SKILLS

LLM & Agents:	LangGraph, LangChain, deepagents, OpenAI Agents SDK, OpenAI & Anthropic APIs, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), hybrid search, tool calling, structured output, prompt engineering & caching
Evaluation:	LLM-as-Judge, LangSmith, evaluation datasets, WER/CER benchmarking, human-annotation consensus (Argilla), Cohen's κ calibration, failure analysis, production QA
ML & Data:	PyTorch, Hugging Face (Transformers, Datasets, Hub), Whisper, fine-tuning data curation, XGBoost, computer vision (Detectron2, YOLO)
Engineering:	Python, TypeScript, FastAPI, Next.js, PostgreSQL, Supabase, Docker, CI/CD, Git, AWS (S3, EC2), Prometheus/Grafana, drift monitoring (Evidently)